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DGPAEP-Net: Deep Gaussian process network with approximate expectation propagation for carbon emission prediction in municipal wastewater treatment processes

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ABSTRACT

Excessive carbon emissions exacerbate climate change and biodiversity loss, making carbon emission monitoring a global environmental focus. Advances in machine learning offer new perspectives for addressing excessive carbon emissions. This study employs an improved Bayesian neural network-based machine learning model to predict carbon emissions, using a full-scale wastewater treatment plant in eastern China as an example. The enhanced model introduces a novel and robust deterministic approximation scheme, leveraging an approximate expectation propagation algorithm and a fully independent training condition approximation to overcome computational and analytical challenges. This model is termed the deep Gaussian process network with approximate expectation propagation (DGPAEP). Before the formal carbon emission prediction, the model's generalization capability was benchmarked against other state-of-the-art Bayesian neural networks on public datasets, ensuring its high performance. Experimental results demonstrate that, compared to other Bayesian neural networks, DGPAEP achieved an increase of 149.07 % in the average test log-likelihood and a 3.05 % reduction in root mean square error. To explore the contribution of different features to the predictive values, Shapley additive explanations (SHAP) were utilized to quantify their impact on the output, thereby better-guiding emission reduction efforts. These findings suggest that machine learning models have the potential to establish a robust framework to enhance the prediction capabilities of carbon emissions in wastewater treatment, offering a promising approach for real-time monitoring and scientific reduction of carbon emissions in wastewater treatment plants.

1. Introduction

The escalation of global warming and recent occurrences of extreme weather have renewed societal focus on carbon emissions (CE) [1,2]. Precise monitoring of these emissions is pivotal for effective carbon reduction strategies [3]. Traditional methods for monitoring CE mainly encompass direct measurement and estimation [4]. Direct measurement utilizes stationary or mobile sampling devices to directly assess emissions from sources, including laboratory testing, necessitating costly equipment and specialized expertise. However, these labor-intensive processes are notably inefficient. In contrast, estimation methods prioritize reasonable ranges over exact measurements, trading off some accuracy for quicker computation and enhanced efficiency [5]. Common estimation strategies encompass carbon emission factor methods and material balance methods [6,7]. Thus, achieving efficient and accurate carbon emission monitoring (CEM) remains a significant challenge in the environmental sector.

The field of artificial intelligence (AI) is advancing rapidly, with machine learning (ML) technologies offering novel insights into reducing CE. Addressing the aforementioned challenges, ML models can integrate the benefits of both direct measurement and estimation methods. These models extract key factors closely correlated with CE, known as "features" in ML. By learning the intrinsic relationships between these features and CE, referred to as "weights" or "probability values", ML models can derive optimal solutions. For instance, recent studies have demonstrated the effectiveness of ensemble learning

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approaches, such as adaptive boosting-long short-term memory (AdaBoost-LSTM), in enhancing the accuracy and robustness of predictions for complex systems like wastewater treatment plants, where precise forecasting of operational parameters can lead to significant energy savings and sustainable practices [8]. Additionally, hybrid models like convolutional neural network-long short-term memory (CNN-LSTM) have shown superior performance in predicting carbon trading prices, highlighting the importance of model robustness and accuracy in practical applications [9]. Furthermore, the application of long short-term memory (LSTM) models for predicting greenhouse gas emissions, such as N_2O, from wastewater treatment plants has demonstrated significant improvements in prediction accuracy and resilience to sudden changes in target sequences, underscoring the potential of ML models in addressing environmental challenges [10]. Crucially, once an ML model is trained, it can predict outcomes for various scenarios based on the relevant features, a capability essential for regression problems in ML [11].

Increasingly, ML technologies are being applied to environmental research [12]. For instance, ML models trained on historical air quality data can predict pollution levels, aiding urban managers in implementing measures to improve air quality [13]. Similarly, ML facilitates ecosystem monitoring, energy management, and environmental pollution source tracking [14]. Inspired by these advancements, this paper focuses on using ML models to predict CE with high accuracy.

ML applications in CE have recently become a research focus. Wan et al. [15] proposed the channel-selection weighted long short-term memory-Gaussian process regression (CSWLSTM-GPR) model for high-precision point and interval predictions in papermaking wastewater treatment systems. This model integrates spatial features from convolutional neural networks (CNN), temporal features from shared weight long short-term memory (SWLSTM), and probabilistic reliability from Gaussian process regression (GPR). While experimental results demonstrate favorable performance, the integrated model's robustness significantly decreases, and its computational complexity rises due to the simple fusion of multiple models, with sub-model compatibility remaining unclear. Chang et al. [16] employed ML and projection pursuit regression (PPR) to assess critical factors influencing carbon dioxide emission forecasts in China. PPR statistical methods were used to handle high-dimensional data and uncover nonlinear structures. Neural networks and fuzzy inference systems were employed to capture complex nonlinear relationships, enhancing prediction accuracy. However, the study's reliance on historical data, which may be biased or incomplete, poses limitations. Additionally, while PPR and ML models handle nonlinear relationships, their complexity and computational costs are high, limiting their broader applicability.

Thus, addressing deficiencies identified in previous literature, this study has made substantial improvements. Carbon emission data from a wastewater treatment plant (WWTP) in Jiangsu province, eastern China, was collected, chosen for its stable and representative nature due to the region's dense population and developed economy. Advanced carbon accounting techniques, as proposed by Li et al. [17], were employed to ensure data reliability. The focus lies on employing ML models capable of capturing complex nonlinear relationships. Gaussian process (GP) models were initially selected, but their traditional forms showed lower accuracy [18]. To enhance accuracy, this study introduced deep Gaussian processes (DGPs) with varying network depths, resembling neural networks with multiple infinitely wide hidden layers. DGPs offer increased flexibility, superior generalization, and improved uncertainty calibration compared to other similar deep models [19]. Additionally, pseudo-data point-based approximation methods were applied to DGPs to reduce computational complexity. This paper introduces a novel model, a Deep Gaussian process network with approximate expectation propagation (DGPAEP), where expectation propagation (EP) is used to approximate the marginal likelihood directly and determine the approximate posterior distribution for inducing outputs.

This study introduces a novel model for predicting CE from a full-scale WWTP. The objectives were: (1) to enhance GP for precise carbon emission predictions at a WWTP, validating DGPAEP performance on public datasets beforehand; (2) to assess the accuracy and reliability of DGPAEP in carbon emission prediction; and (3) to utilize Shapley additive explanations (SHAP) values to determine the significance of input features, guiding targeted emission reduction efforts for plant operators.

2. Model's fundamental architecture

2.1. Deep Gaussian processes

DGPs are multilayer hierarchical generalizations of GPs and are formally equivalent to neural networks with multiple, infinitely wide hidden layers [20]. Assuming a training set consisting of N input and observation pairs in D dimensions, represented as $(\mathbf{x}_n, y_n)$, where the outputs are assumed to be real-valued scalars for simplicity but other data types can be accommodated, the probabilistic model of a DGP with L layers can be expressed as follows (Eqs. (1)-(3)),

$$p(f_l|\Theta_l) = \mathcal{GP}(f_l; \mathbf{0}, \mathbf{K}_l), l = 1, \cdots, L \tag{1}$$

$$p\left(\mathbf{h}_l|f_l, \mathbf{h}_{l-1}, \sigma_l^2\right) = \prod_n \mathcal{N}\left(h_{l,n}; f_l\left(h_{l-1,n}\right), \sigma_l^2\right), h_{1,n} = \mathbf{x}_n \tag{2}$$

$$p\left(\mathbf{y}|f_L, \mathbf{h}_{L-1}, \sigma_L^2\right) = \prod_n \mathcal{N}\left(y_n; f_L\left(h_{L-1,n}\right), \sigma_L^2\right) \tag{3}$$

where hidden layers are denoted $h_{l,n}$and the functions in each layer, f_l. In a more formal context, introduce a zero mean GP before the mapping f_l. Specifically, when provided with the inputs for f_l, any finite set of function values follows a multivariate Gaussian distribution under the prior, denoted as $p(\mathbf{f}_l) = \mathcal{N}(\mathbf{f}; \mathbf{0}, \mathbf{K_{ff}})$. It's important to note that these function values and, consequently, the hidden variables are not marginally normally distributed, given that the inputs are random variables. The model reverts to Gaussian process regression when $L = 1$. Additionally, when the inputs $\{\mathbf{x}_n\}$ are both unknown and random, the model transforms into a DGP latent variable model. An illustration of DGPs is presented in Fig. 1 for the scenario where $L = 2$ and $\dim(h_1) = 2$ [21]. This network, along with the suggested approximation and training algorithm, is employed to model a value function for the mountain car problem based on a limited number of noisy evaluations.

2.2. Deep Gaussian process network with approximate expectation propagation

The overall schematic diagram of the DGPAEP framework is depicted in Fig. 2. This model is essentially an extension of the Gaussian neural network, structurally resembling an artificial neural network (ANN). Similar to ANN, it consists of an input layer, hidden layers, and an output layer. However, unlike traditional neural networks, the transmission between the hidden layers does not rely on weights but rather utilizes the mean and variance derived from Gaussian distributions. By iteratively updating the mean and variance, and appropriately adjusting based on the loss function's value, the model continues to optimize until the loss function reaches its expected minimum, at which point the final predictive result is output, achieving the desired prediction.

The length-scales in the Gaussian process layers control the smoothness of the function modeled by the kernel. Specifically, a smaller length-scale indicates that the function varies more rapidly, while a larger length-scale suggests a smoother function. In the context of carbon emission prediction, the length-scales help the model capture the varying influence of different features (e.g., chemical dosage, electricity consumption) on emissions. The inducing inputs are a set of pseudo-data points used to approximate the full Gaussian process, reducing computational complexity. These points act as a sparse representation of the

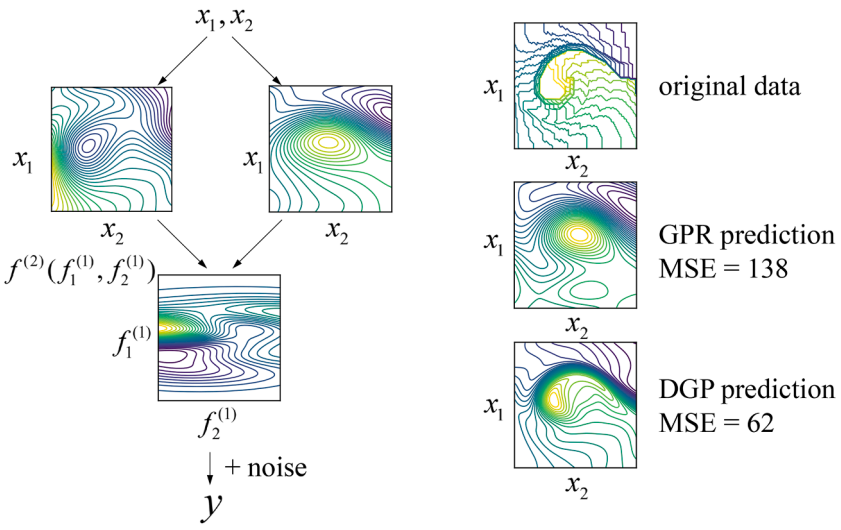


Fig. 1. An instance of a DGP featuring two GP layers and one 2-D hidden layer. The left graphs show latent functions in each layer, two functions in the first layer and one in the second layer, learnt by using the proposed approach. The right graph shows the training data [top] and the predictions of the overall function mapping from inputs to outputs made by a Gaussian process regression (GPR) [middle] and the DGP on the left [bottom].

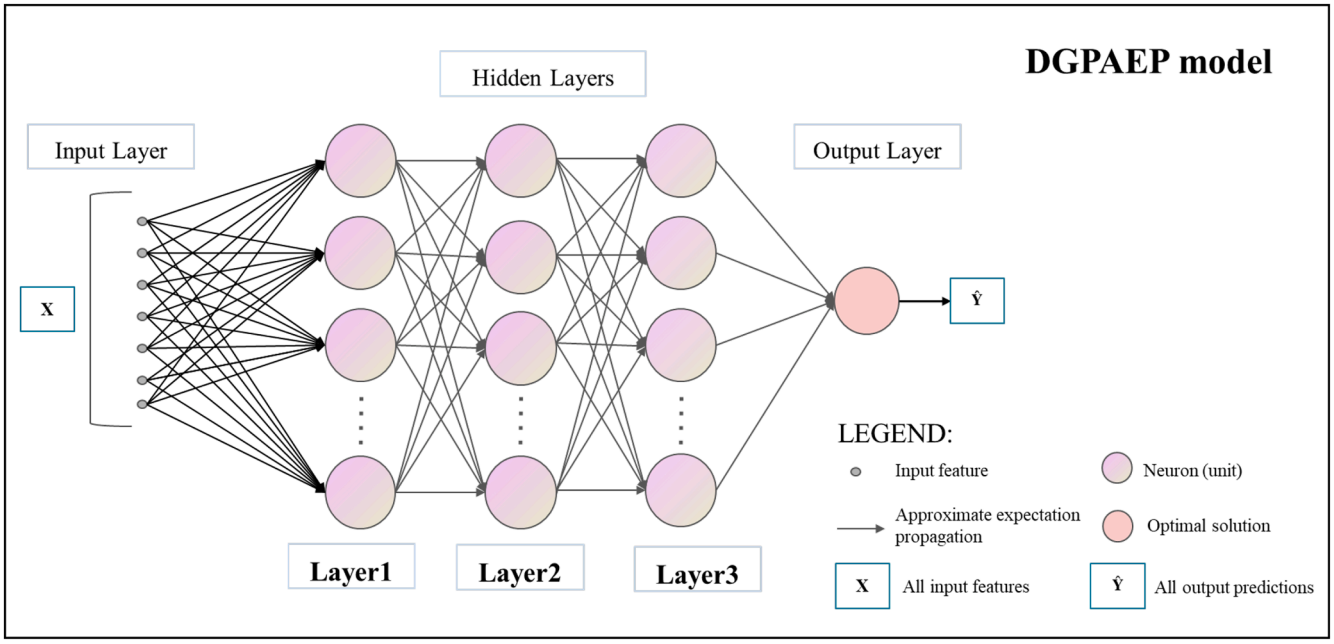


Fig. 2. Schematic diagram of the DGPAEP network model.

data, allowing the model to focus on the most informative regions of the input space. By optimizing the locations of these inducing inputs, the model can better approximate the true posterior distribution, improving both computational efficiency and predictive accuracy.

Fig. 3 provides a detailed explanation of the DGPAEP learning process. The training process strictly follows the sequence and principles outlined in Fig. 1, visualizing each step to explain the overall implementation of the proposed DGPAEP model. The first part of the model is the input layer. Using the biological treatment stage of the wastewater treatment process as an example, this stage comprises 18 features, each exhibiting random and uncertain distributions, as illustrated by the scatter plot. Based on the raw data, the Gaussian distribution for each feature is computed, which is critical for the Gaussian neural network. Through these calculations, the mean and variance of each feature are obtained, allowing the Gaussian distribution curve and the data distribution curve to be fitted. The second part is the core of this network. From the first part, the Gaussian distribution for each feature is derived. To visualize the Gaussian distribution of each feature, normalization was applied to address differences in magnitude. The figure clearly displays the Gaussian distribution for each feature. Subsequently, approximate posterior distributions are computed using approximate inference methods. To reduce the model's computational complexity, the fully independent training conditional (FITC) model was employed, ultimately completing the Gaussian fitting process. The third part is the output layer, where the trained model outputs the optimal predictive values. The underlying principles of each part are described in detail below.

2.2.1. *Fully independent training conditional approximation*

To address the prohibitive cubic increase in computational complexity associated with full GP models concerning the number of training instances, sparse approximation techniques are applied [22].

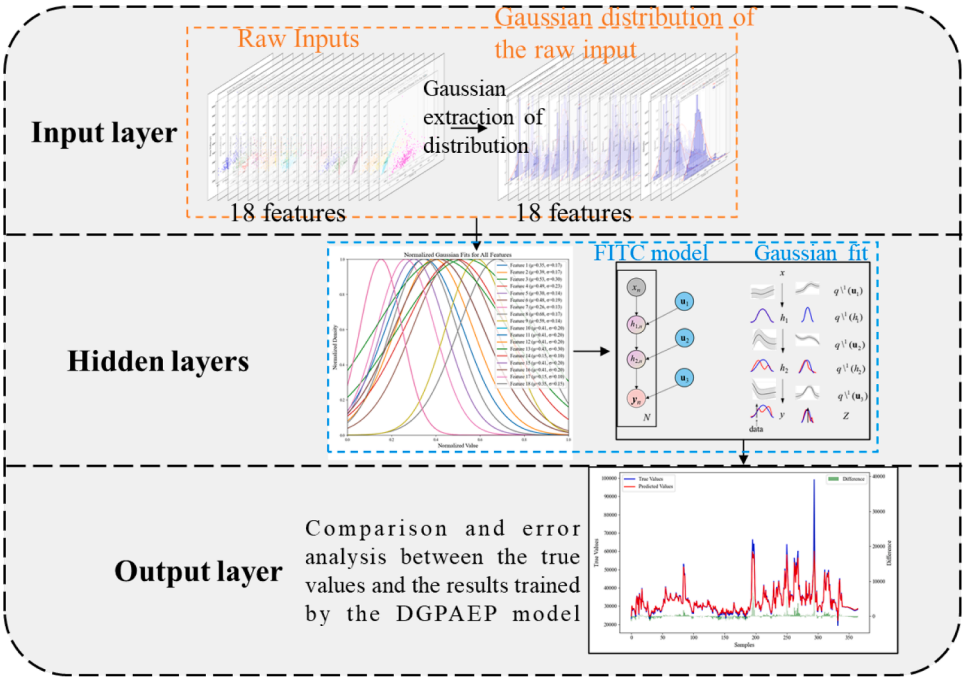


Fig. 3. Explanation of the DGPAEP framework. The meaning of each sub-figure is described at the top of the image. The numbers represent the number of features in the input layer. In the output results, blue indicates the true values, red denotes the predicted values, and the green bar chart represents the absolute error between the true and predicted values.

The FITC method is employed in this context to mitigate the computational challenges [23]. The FITC approximation model can be written as follows (Eqs. (4)-(6)),

$$p(\mathbf{u}_l|\Theta_l) = \mathcal{N}\left(\mathbf{u}_l; \mathbf{0}, \mathbf{K}_{\mathbf{u}_{l-1}}, \mathbf{u}_{l-1}\right), l = 1, \cdots, L \tag{4}$$

$$p\left(\mathbf{h}_l|\mathbf{u}_l, \mathbf{h}_{l-1}, \sigma_l^2\right) = \prod_n \mathcal{N}\left(h_{l,n}; \mathbf{C}_{n,l}\mathbf{u}_l, \mathbf{R}_{n,l}\right) \tag{5}$$

$$p\left(\mathbf{y}|\mathbf{u}_L, \mathbf{H}_{L-1}, \sigma_L^2\right) = \prod_n \mathcal{N}\left(y_n; \mathbf{C}_{n,L}\mathbf{u}_L, \mathbf{R}_{n,L}\right) \tag{6}$$

where $\mathbf{C}_{n,l} = \mathbf{K}_{\mathbf{h}_{l,n},\mathbf{u}_l}\mathbf{K}_{\mathbf{u}_l,\mathbf{u}_l}^{-1}$ and $\mathbf{R}_{n,l} = \mathbf{K}_{\mathbf{h}_{l,n},\mathbf{h}_{l,n}} - \mathbf{K}_{\mathbf{h}_{l,n},\mathbf{u}_l}\mathbf{K}_{\mathbf{u}_l,\mathbf{u}_l}^{-1}\mathbf{K}_{\mathbf{u}_l,\mathbf{h}_{l,n}} + \sigma_l^2 I$. The set $\mathbf{u}$ mentioned in the above formula is often denoted as the set of inducing outputs or pseudo targets, and their input locations, $\mathbf{z}$, can be chosen by optimizing the approximate marginal likelihood. The graphical model is shown in Fig. 4.

2.2.2. Approximate Bayesian inference via EP

Building upon the preceding section, the focus now shifts to the inference problem concerning inducing outputs $\mathbf{u} = \{\mathbf{u}_l\}_{l=1}^L$ and model parameters$\alpha = \{\mathbf{z}_l, \Theta_l\}_{l=1}^L$. The posterior distribution of inducing outputs, written as$p(\mathbf{u}|\mathbf{X},\mathbf{y}) \propto p(\mathbf{u})\prod_n p\left(y_n|\mathbf{u},\mathbf{X}_n\right)$, is a key consideration. Given a test input $\mathbf{x}^*$, the associated prediction is expressed as$p(y^*|\mathbf{x}^*,\mathbf{X},\mathbf{y}) = \int \mathbf{du} p(\mathbf{u}|\mathbf{X},\mathbf{y}) p(y^*|\mathbf{u},\mathbf{x}^*)$. However, solving for these values proves to be a formidable task within the domain of DGPs. To tackle this challenge, EP

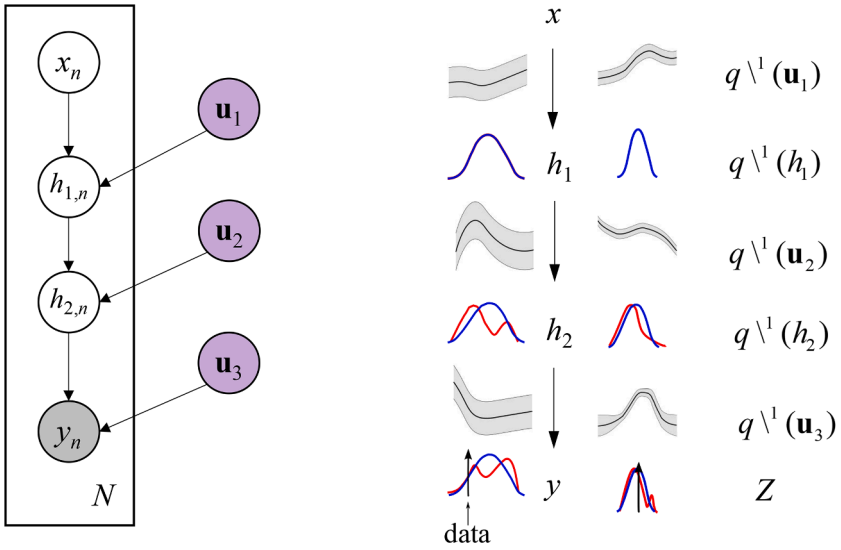


Fig. 4. The graphical model of proposed DGP-FITC model. The diagram on the left illustrates the implementation mechanism of the model, while the one on the right vividly depicts the process of training and fitting the data.

energy function with a tied factor emerges as a pivotal technique.

EP approximates the marginal likelihood by iteratively refining approximate data factors $\{\widetilde{t}_n(\mathbf{u})\}_{n=1}^N$, which capture the contribution of each datapoint to the posterior. The approximate posterior is given by $q(\mathbf{u})\propto p(\mathbf{u})\prod_n \widetilde{t}_n(\mathbf{u})$, where each factor $\widetilde{t}_n(\mathbf{u})$ is an unnormalized Gaussian. The EP algorithm minimizes the Kullback-Leibler (KL) divergence between the cavity distribution $q^{\setminus n}(\mathbf{u})\propto q(\mathbf{u})/\widetilde{t}_n(\mathbf{u})$ and the true posterior $p(y_n|\mathbf{u}, \mathbf{X}_n)$, updating the factors iteratively until convergence. This process provides an approximation to the marginal likelihood (Eq. (7)):

$$\log p(y|\alpha) \approx \mathscr{F}(\alpha) = \phi(\theta) - \phi(\theta_{\text{prior}}) + \sum_{n=1}^{N}\left[\log \mathscr{Z}_n + \phi(\theta^{\setminus 1}) - \phi(\theta)\right] \quad (7)$$

where θ,θ_{prior} and $\theta^{\setminus 1}$ are the natural parameters of the approximate posterior $q(\mathbf{u})$, the cavity distribution $q^{\setminus n}(\mathbf{u})$, and the prior $p(\mathbf{u})$, respectively, and $\phi(\cdot)$ is the log normalizer of a Gaussian distribution.

To reduce the memory cost of EP, which scales as $\mathrm{O}(NLM^2)$, where N is the number of datapoints, L is the number of layers, and M is the number of pseudo datapoints, a tied factor constraint is employed, leading to the stochastic expectation propagation (SEP) algorithm. In SEP, the approximate posterior is simplified to $q(\mathbf{u})\propto p(\mathbf{u})g(\mathbf{u})^N$, where $g(\mathbf{u})$ is an average data factor. This reduces the memory requirement to $\mathrm{O}(LM^2)$ while maintaining comparable performance to EP. The SEP algorithm iteratively updates the average factor $g(\mathbf{u})$ by minimizing the KL divergence between the cavity distribution $q^{\setminus 1}(\mathbf{u})\propto q(\mathbf{u})/g(\mathbf{u})$ and the true posterior.

It was found that directly optimizing the EP energy function with tied factors outperforms SEP in terms of computational efficiency and scalability. The energy function simplifies to (Eq. (8)):

$$\mathscr{F}(\alpha) = (1-N)\phi(\theta) + N\phi(\theta^{\setminus 1}) - \phi(\theta_{\text{prior}}) + \sum_{n=1}^{N}\log \mathscr{Z}_n \quad (8)$$

where $\theta = \theta_{\text{prior}} + N\theta_1$ and $\theta^{\setminus 1} = \theta_{\text{prior}} + (N-1)\theta_1$. This approach eliminates the need for a double-loop algorithm, enabling joint optimization of hyperparameters and inducing inputs. However, it is important to note that this method averages moments of $q^{\setminus 1}(\mathbf{u})p(y_n|\mathbf{u}, \mathbf{x}_n)$ over datapoints, whereas EP averages natural parameters, which is more principled but less tractable.

The advantages of EP over other inference methods include its ability to provide accurate approximations to the posterior and marginal likelihood, its scalability through tied factors, and its flexibility in handling complex models like DGPs. By leveraging EP, a balance between computational efficiency and inference accuracy is achieved, making it well-suited for large-scale applications.

In the DGPAEP framework, EP is then applied to further refine the posterior distribution of the inducing outputs $\mathbf{u}$. While FITC provides an efficient sparse approximation by introducing inducing outputs, EP plays a key role in refining the approximation by updating the posterior distribution of these inducing outputs. FITC simplifies the model by introducing a sparsified structure, reducing the original complexity of full GP inference to $\mathrm{O}(LM^2)$ The relationship between FITC and EP can be described as follows: FITC is initially applied to sparse the model by introducing a small set of inducing outputs. This simplification reduces computational overhead, making it feasible to apply further inference methods such as EP. Once the sparse approximation is made, EP is used to refine the posterior distribution of the inducing outputs. EP optimizes the parameters of the posterior.

2.2.3. Probabilistic backpropagation for deep Gaussian processes

Following the completion of the new energy computation, the calculation of $\log \mathscr{Z}_n = \log \int \mathrm{d}\mathbf{u} q^{\setminus 1}(\mathbf{u})p(y_n|\mathbf{u}, \mathbf{x}_n)$ remains pending, and this matter will be addressed in the subsequent steps. The computation of $\log \mathscr{Z}_n$ in the aforementioned objective function becomes analytically challenging for $L \geq 1$ due to the nonlinearity of the likelihood given the inducing outputs $\mathbf{u}$ and the intricate distribution resulting from the Gaussian cavity propagation through each layer. Nonetheless, for specific choices of covariance functions $\{\mathbf{K}_l\}_{l=1}^L$, it is feasible to employ an efficient and accurate approximation. This approximation involves propagating a Gaussian through the initial layer of the network, projecting the non-Gaussian distribution back to a moment-matched Gaussian, and iterating the process through subsequent layers. This approach shares algorithmic similarity with nesting assumed density filtering (ADF) and constitutes a pivotal component of the probabilistic backpropagation algorithm, commonly applied to standard neural networks. By reintroducing the hidden variables in the middle layers, we perform a Gaussian approximation to $\mathscr{Z}$ in a sequential fashion, as illustrated in Fig. 4.

The analytical tractability of the mean and variance of the Gaussian approximation in each intermediate layer allows for explicit computation of their gradients to the mean and variance of the input distribution, as well as the parameters of the current layers. To fulfill the need for gradients related to the $\log \mathscr{Z}$ approximation, it is imperative to retain these outcomes during forward propagation. Subsequently, in the output layer, both the approximate $\log \mathscr{Z}$ and its gradients are computed. The chain rule is then applied in the backward step, facilitating differentiation through the ADF procedure. This procedural parallelism with the backpropagation algorithm in standard parametric neural networks begets the nomenclature "probabilistic backpropagation" [24].

2.3. Model evaluation metrics

Predictive performance on the test set is evaluated using some widely used metrics like root mean squared error (RMSE), mean log-likelihood (MLL), mean absolute error (MAE) and coefficient of determination R^2. This study, drawing on relevant literature [25,26], comprehensively considers the characteristics of performance metrics in regression problems and selects RMSE to measure the error between predicted and actual values (Eq. (9)) and MLL to assess the model's generalization capability (Eq. (10)). MLL is an indicator used in statistics and ML to evaluate the goodness of fit of a model, with higher MLL values indicating better model performance. In these equations, n represents the sample size, y_i is the i th actual value, m denotes the mean of the model's predictions, and v signifies the variance of the model's predictions. $-\frac{1}{2}\log(2\pi v)$ represents the log-likelihood part for the variance v, $-\frac{1}{2}\frac{(y_i-m)^2}{v}$ accounts for the log-likelihood part related to the error between actual and predicted values.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2} \quad (9)$$

$$MLL = \frac{1}{n}\sum_{i=1}^{n}\left(-\frac{1}{2}\log(2\pi v) - \frac{1}{2}\frac{(y_i-m)^2}{v}\right) \quad (10)$$

3. Experimental data and training parameter setting

3.1. Data acquisition and description

In this study, the data was sourced from a WWTP in Jiangsu province, China. The data holds credibility primarily based on the following criteria: 1) The objectivity and representativeness of the data source, stemming from an economically developed region with higher demands for sewage treatment, ensuring the collection of long-term stable data. 2) The presence of air pollution monitoring stations within the study area. 3) Selection of sites with a multitude of measured carbon emission parameters, including chemical, biological, and some physical parameters. Based on these criteria, this data was chosen as the case study for the research. Fig. 5 is presented as a flowchart illustrating the specific

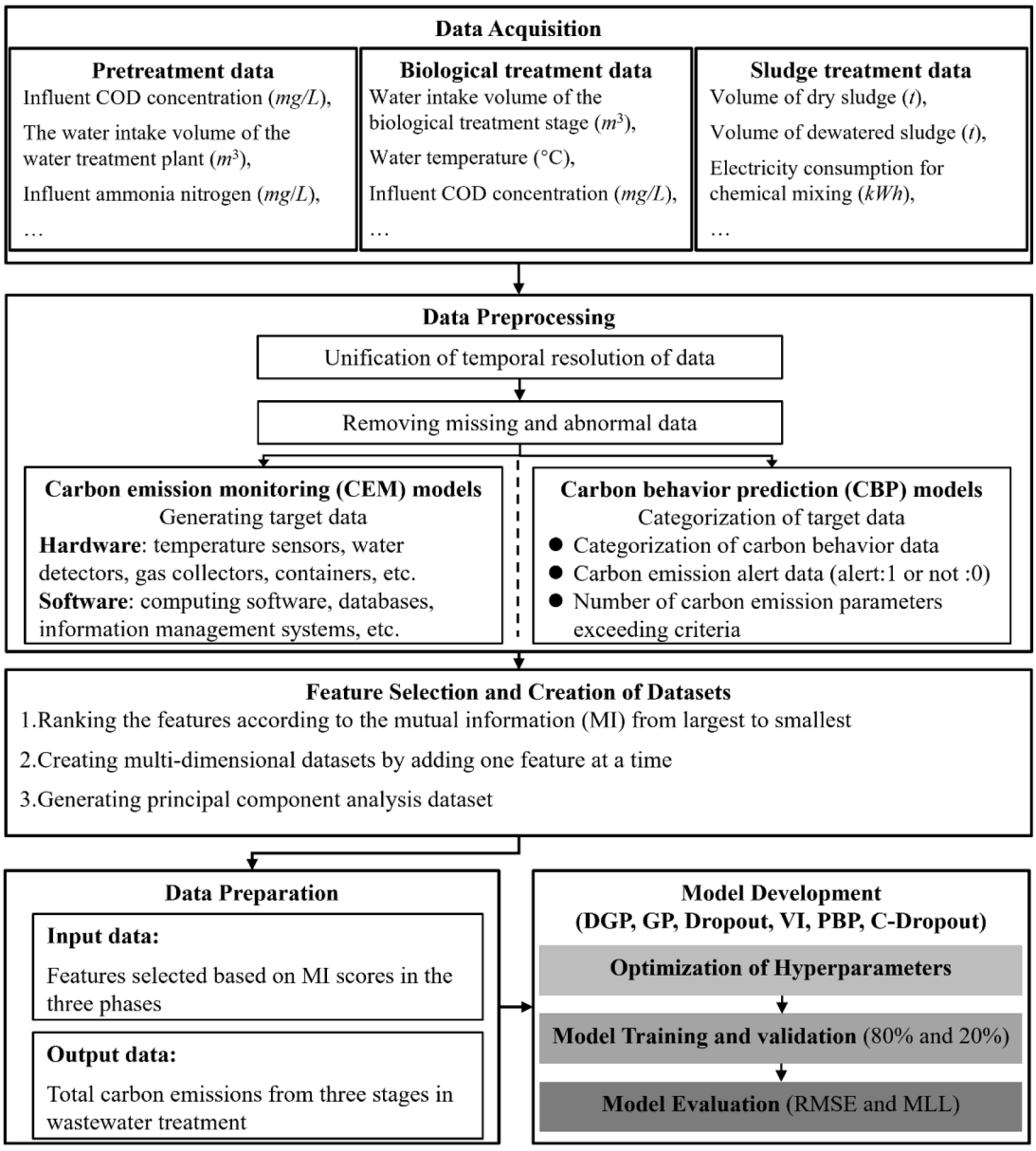


Fig. 5. Logical flow of applying machine learning models to predict carbon emissions in a full-scale WWTP.

experimental process of this paper, encompassing the entire procedure from data acquisition to model training.

In the pretreatment stage, we included the CO_2 emissions generated by the electricity consumption of the off-site pump in the accounting. This is not typically included in traditional accounting methods, but the carbon emissions from this electricity consumption cannot be ignored. Therefore, we incorporated this carbon source to improve the accuracy of the calculation results. The measurement of emissions was performed using a combination of direct measurement techniques and established calculation methods. Gas analyzers were used to directly measure the concentration of CO_2, CH_4, and N_2O at various points in the treatment process, particularly in the aeration tanks and anaerobic digesters. Flow meters were also installed at key points, such as influent and effluent entry points and throughout the aeration and digestion processes, to measure the flow rates of water and gases, which was crucial for accurately quantifying the volume of gases emitted.

Fossil-derived CO_2 Emissions: The direct CO_2 emissions from fossil sources were calculated using the following equation (Eq. (11)):

$$m_{CO_2f1,i} = \left[Q_{0i} \times \left(COD_{0i} - COD_{1i} \times 10^{-3}\right] \times k_f \quad (11)$$

where $m_{CO_2f1,i}$ is the direct CO_2 emission from fossil carbon on day i, Q_{0i} is the influent rate, COD_{0i} and COD_{1i} are the influent and effluent COD concentrations, respectively, and k_f is the proportion of fossil carbon in the total organic carbon (TOC) in sewage, taken as 8.75 % based on previous studies [27–29].

Indirect CO_2 Emissions: The indirect CO_2 emissions from electricity consumption were calculated using the following equation (Eq. (12)):

$$m_{e,j,i} = W_{j,i} \times EF_e \quad (12)$$

where $m_{e,j,i}$ is the CO_2 emission from electricity consumption in link j on day i, $W_{j,i}$ is the purchased electricity used for operation in link j, and EF_e is the carbon emission factor of electricity, taken as 0.6299 kg CO_2/(kW·h) based on local data from Jiangsu SinoCarbon Innovation & Investment Co., Ltd.

Chemical consumption emissions: The CO_2 emissions from chemical consumption were calculated using the following equation (Eq. (13)):

$$m_{p1,i} = \sum_{l=1}^{n} \left(m_{l,i} \times EF_l \right) \tag{13}$$

where $m_{p1,i}$ is the CO_2 emission from chemical consumption on day i, $m_{l,i}$ is the mass of the l-th chemical used, and $EF_l$is the emission factor of the l-th chemical. The emission factors for chemicals such as polyacrylamide (PAM), $FeCl_3$, and other flocculants were obtained from literature [30, 31].

The emission factors for electricity and chemicals were sourced from reliable references, ensuring the accuracy of the calculations. For example, the emission factor for electricity was taken as 0.6299 kg CO_2/(kW·h), while emission factors for chemicals such as PAM (1.50 kg CO_2/kg) and $FeCl_3$ (2.71 kg CO_2/kg) were obtained from prior studies. These values were crucial for quantifying the overall carbon footprint of the pretreatment stage. The results indicate that direct emissions in the pretreatment stage amount to 355 tCO_2/year, while indirect emissions are significantly higher, totaling 6405 tCO_2/year. This demonstrates that indirect emissions are approximately 18 times higher than direct emissions, emphasizing the need for greater focus on reducing indirect emissions. Among the indirect emissions, electricity consumption accounts for 2050t CO_2/year, while chemical consumption contributes 4354 tCO_2/year, making chemical-related emissions the primary target for reduction efforts. In terms of emission intensity, the data shows a strong linear relationship between emissions and intensity, with the lowest emission intensity observed for PAM. Sources such as fossil-derived CO_2, aeration, and chemical mixers, as well as primary sedimentation tanks, all have emission intensities below 0.01 kg CO_2/m^3, suggesting that these sources are less critical for emission reduction. However, the carbon emissions generated by water pumps, particularly the off-site pump, require attention. In conclusion, reducing carbon emissions in the pretreatment stage should primarily focus on optimizing chemical usage. Strategies such as implementing intelligent chemical dosing systems and selecting chemicals with lower emission factors could significantly reduce emissions while maintaining treatment efficiency.

In typical studies, CE are calculated starting from the point when wastewater enters the treatment plant [32]. It is asserted that the electricity consumed by pumps to draw water from the municipal network before it enters the treatment plant, resulting in indirect CO_2 emissions, is inevitable and should be included within the scope of CE. "Carbon" in this context refers to CO_2, CH_4, and N_2O. The case study in this paper divides the internal treatment stages into three parts: pretreatment, biological treatment, and sludge treatment stages.

In this case study, the stage from the municipal network to the primary sedimentation tank is termed the pretreatment phase. Specific sources of CE in this phase include indirect CO_2 emissions from pumps outside the municipal network, sewage lifting pumps, and aeration sand settling tanks. Organic matter in the aeration sand settling tank decomposes during aeration and settling processes, directly generating fossil-source CO_2. Additionally, indirect CO_2 emissions arise from chemical additives in the primary sedimentation tank. Indirect emissions in the pretreatment phase are primarily attributed to electricity and chemical consumption. During the biological treatment phase, significant direct emissions such as CH_4 and N_2O are prevalent in wastewater treatment. In the biological treatment phase, CE originate from CH_4 generated during anaerobic digestion, N_2O produced in anaerobic denitrification, fossil-source CO_2 used for denitrification reactions, indirect CO_2 sources, N_2O from incomplete nitrification and nitritation reactions, organic matter decomposition containing fossil carbon, indirect CO_2 from electricity consumption in aeration tanks, and chemical consumption in secondary settling and disinfection tanks. Anaerobic digestion is the primary stage for CH_4 production, critical for achieving energy neutrality in wastewater treatment plants. Furthermore, the global warming potential (GWP) of CH_4, 28 times that of CO_2, highlights the significance of accurately measuring CH_4 emissions in data [33]. N_2O primarily forms as an intermediate product during anoxic denitrification processes. Accumulation of NH_2OH occurs under low oxygen conditions in the anoxic phase, oxidizing to N_2O catalyzed by hydroxylamine oxidoreductase enzymes. Despite its low emission volume, N_2O has a potent GWP of 298 times that of CO_2, necessitating careful consideration [34]. Aeration processes decompose organic matter in wastewater, releasing fossil-source CO_2. In the biological treatment phase, aeration constitutes a major source of indirect CO_2 emissions due to various chemical reactions under aerobic conditions. Microbial decomposition of organic nitrogen compounds into NH_4^+-N precedes nitrification, potentially generating N_2O as a byproduct under oxygen-limited conditions [35]. Phosphorus-accumulating bacteria become active under aerobic conditions, absorbing excess phosphorus from wastewater and removing it as phosphate. Thus, aeration processes consume substantial electricity, resulting in the highest indirect CO_2 emissions. Given the low-carbon status of urban sewage in China, additional carbon sources are required to enhance anaerobic denitrification processes and maintain a specified C/N ratio, ultimately emitted as CO_2. Following biological treatment, wastewater requires further sedimentation and disinfection before discharge, with chemical additions contributing to indirect emissions during this process. Sludge treatment is an indispensable component of wastewater treatment plants. Residual sludge from secondary settling tanks typically retains a moisture content as high as 80 %. To reduce moisture content, centrifuges are commonly employed for sludge concentration, a process that generates indirect CO_2 emissions [36]. Additionally, polyacrylamide (PAM) is often added to enhance the separation of sludge and water, contributing to further indirect CO_2 emissions due to chemical consumption. Subsequently, sludge undergoes a drying process where heat, usually generated by heat pumps, is required, thereby also emitting indirect CO_2. Furthermore, thermal energy is directly produced via coal combustion in wastewater treatment plants. Dried sludge can undergo anaerobic fermentation to further recover CH_4, which can be utilized for electricity generation or heating to offset some CE. Transporting sludge offsite via trucks powered by diesel also results in indirect CO_2 emissions. In summary, direct emissions at this stage stem from CH_4 produced during the anaerobic digestion of sludge. Indirect CE primarily result from sludge concentration, drying processes, chemical usage, and electricity consumption from diesel fuel.

3.2. Analysis of the carbon dioxide emission dataset from the wastewater treatment plant

The entire dataset is sourced from a real-world WWTP and is divided into three stages: pretreatment, biological treatment, and sludge treatment. The data resolution in this study is 365 × 45, with each stage encompassing data from the entire year of 2021, totaling 45 different types of annual data across the three stages. A summary of these data types is provided in Table 1, including the units and specific metrics recorded for all 365 days. These data types represent various parameters involved in the wastewater treatment process, including energy consumption, chemical usage, and various performance metrics for each treatment stage. Table 1 provides a complete list of the 45 data types used in this study, ensuring a comprehensive overview of the dataset's scope and complexity. The data will undergo feature selection to identify the most relevant variables for model training.

Within the carbon emission data, it has been categorized into three stages. To better illustrate the superiority of DGPAEP in this paper on this data, the dataset utilized in this experiment has also been classified into three categories. Fig. 6 visually depicts the overall situation of total emissions across the three stages. To facilitate a clearer presentation of trends and fluctuations, normalization has been applied due to the substantial variations in CE across different stages. This process compresses the emission values into the [0,1] interval. "Carbon" in this context encompasses CO_2, CH_4, and N_2O. Each stage exhibits distinct

Table 1
Summary of 45 data types collected for the WWTP study with 365 samples.

Phase	Number	Data type
Pretreatment stage	1	Water Treatment Plant Inflow (WTPI)
	2	Influent COD Concentration (ICC)
	3	COD Concentration in Effluent from Aerated Sand Settling tank (COD-ASS-E)
	4	Electricity Consumption of External Pump (ECEP)
	5	Electricity Consumption of Sewage Lift Pump (ECSLP)
	6	Electricity Consumption of Aerated Sand Settling Tank (ECASST)
	7	Electricity Consumption of Chemical Mixing (ECCM)
	8	Electricity Consumption of Primary Sedimentation Tank (ECPST)
	9	PAM Dosage in Primary Sedimentation Tank (PAM-PST)
	10	Dosage of Ferric Sulfate (Fe2(SO4)3) Polymer in Primary Sedimentation Tank (Fe2(SO4)3-PST)
	11	Dosage of Ferric Chloride (FeCl3) in Primary Sedimentation Tank (FeCl3-PST)
Biological treatment stage	12	Influent Flow (IF)
	13	Effluent Flow (EF)
	14	Water Temperature (WT)
	15	Influent COD Concentration (ICC)
	16	Effluent COD Concentration (ECC)
	17	Total Sludge Volume (TSV)
	18	Sludge Organic Load (SOL)
	19	Influent TN Concentration (ITNC)
	20	Effluent TN Concentration (ETNC)
	21	Electricity Consumption of Aerobic Aeration (ECAE)
	22	Electricity Consumption of Anaerobic Digestion (ECAD)
	23	Electricity Consumption of Chemical Mixing (ECCM)
	24	PAM Dosage in Secondary Sedimentation (PAM-SS)
	25	Ferric Sulfate (Fe2(SO4)3) Polymer Dosage in Secondary Sedimentation (Fe2(SO4)3-SS)
	26	Electricity Consumption of Secondary Sedimentation Tank (ECSST)
	27	Electricity Consumption of Disinfection Tank (ECDT)
	28	Ferric Chloride (FeCl3) Dosage in Secondary Sedimentation (FeCl3-SS)
	29	Sodium Hypochlorite (NaClO) Dosage in Disinfection Tank (NaClO-DT)
Sludge treatment stage	30	Total Sludge Volume (TSV)
	31	Sludge Organic Load (SOL)
	32	Dewatered Sludge Quantity (DSQ)
	33	Electricity Consumption of Sludge Thickening (ECST)
	34	Electricity Consumption of Anaerobic Sludge Digestion (ECASD)
	35	Electricity Consumption of Chemical Mixing (ECCM)
	36	Electricity Consumption of Sludge Drying Heat Pump (ECSDHP)
	37	PAM Dosage in Sludge Thickening (PAM-ST)
	38	Coal or Natural Gas Consumption for Sludge Drying (CNGCSD)
	39	CH_4 Recovery Volume from Sludge Digestion (CRVSD)
	40	Transport Distance (TD)
	41	Annual Fuel Consumption of Sludge Transport Trucks (AFCSTT)
Other stage	42	Electricity Consumption in Other Processes (ECOP)
	43	Chemical 1 Consumption in Other Processes (CCOP1)
	44	Chemical 2 Consumption in Other Processes (CCOP2)
	45	Chemical 3 Consumption in Other Processes (CCOP3)

carbon emission characteristics. For instance, the pre-treatment phase is predominantly characterized by indirect CO_2 emissions, as it requires electricity for pumping and chemical agents for precipitating suspended solids. During the biological treatment phase, biological reactions directly result in emissions of CH_4 and N_2O. In the sludge treatment phase, electricity and heat are utilized for sludge concentration and drying, predominantly generating indirect CO_2 emissions. From the graph, it is evident that total CE exhibit significant fluctuations across all three stages, indicating one of the characteristics of CE is their pronounced uncertainty.

3.3. Data feature selection

Common feature selection algorithms include recursive feature elimination (RFE), Fisher score, and Laplacian score. Among these, RFE employs an iterative training process, where one feature is removed in each iteration to achieve dimensionality reduction. However, its stability is constrained by the classifier used, which limits its development. The Fisher score method operates on the principle that features possess discriminative power, thereby ranking them and selecting those with stronger discriminative capabilities. Nevertheless, this method faces challenges in threshold setting, overemphasizes features with class information, and neglects the preservation of local features. The Laplacian score method calculates a score for each individual feature, selecting those with higher variance and the most intact local information. However, its drawback lies in its poor sensitivity to feature weights. The RReliefF regression algorithm, on the other hand, ranks features by weighting them based on the nearest neighbors of the response data, demonstrating strong sensitivity to features. RReliefF is a feature selector applicable to multi-category variables, which calculates weights by penalizing or rewarding neighboring data points [37]. A higher RReliefF value indicates a stronger association between the feature and the target variable. In this study, the RReliefF values for the data presented in Table 1 were computed, and the results are illustrated in the Fig. 7, where the numbering aligns with that in the table.

3.4. Training parameter settings

In all experiments presented in this study, our objective function is optimized using Adam with the default learning rate, which is set to 0.02 [38]. The same hyperparameters (such as batch size, weight decay, etc.) were used for all models in this study. These hyperparameters were determined based on previous research and preliminary experiments, ensuring a consistent and fair comparison across all models. No further hyperparameter tuning, such as grid search or cross-validation, was performed in this study.

For each layer, an exponentiated quadratic kernel (also known as the radial basis function, or RBF kernel) with automatic relevance determination (ARD) length-scales is employed. The exponentiated quadratic kernel is chosen for its ability to model smooth, continuous functions, which is particularly well-suited for capturing the complex, nonlinear relationships often present in environmental and engineering datasets. Its mathematical form, $k(\mathbf{x},\mathbf{x}') = \sigma_f^2 \exp\left(-\frac{1}{2}\sum_{d=1}^{D}\frac{(x_d - x')^2}{l_d^2}\right)$, ensures that the kernel is infinitely differentiable, making it ideal for tasks requiring high precision in function approximation. The use of ARD length-scales allows the model to automatically learn the importance of each input dimension by assigning a separate length-scale parameter l_d to each dimension d. This is particularly advantageous in high-dimensional datasets, where not all input features may contribute equally to the output. By learning these length-scales, ARD effectively performs feature selection, improving the model's interpretability and predictive performance. For example, in the context of carbon emission prediction, ARD can identify which operational parameters are most influential, thereby enhancing the model's ability to generalize to new data.

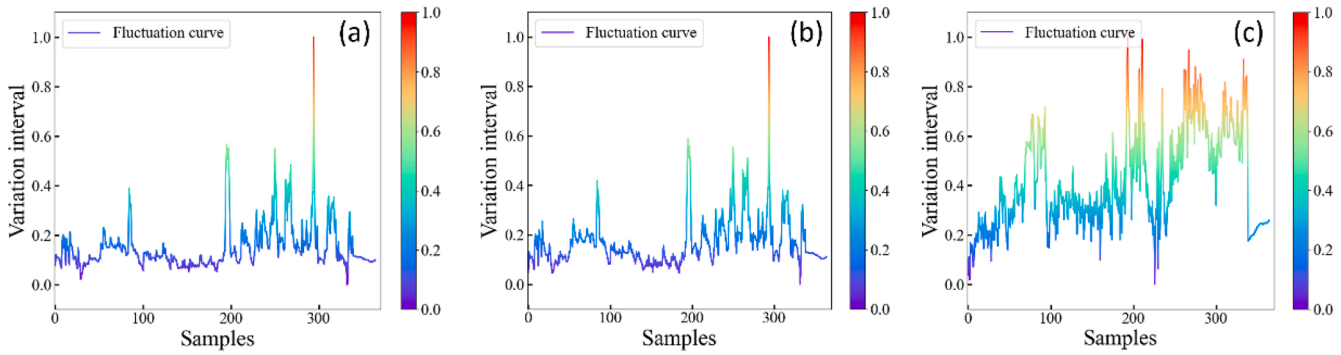


Fig. 6. The variability in the total carbon emissions across the three stages is demonstrated. (a) Pretreatment stage, (b) Biological treatment stage, (c) Sludge treatment stage.

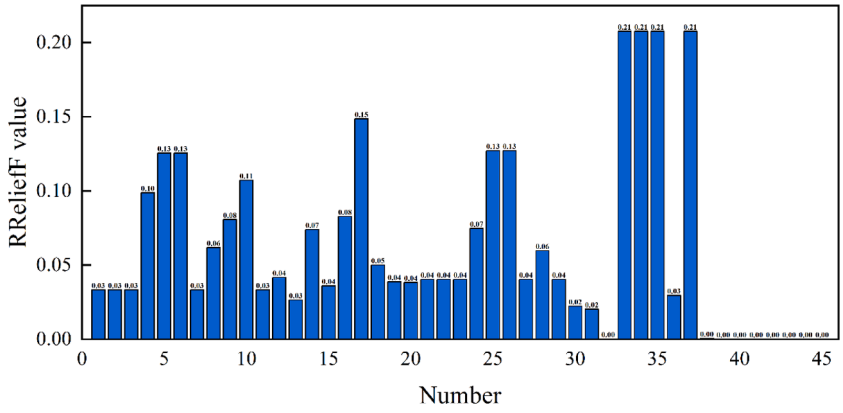


Fig. 7. Feature selection is implemented using the RReliefF regression algorithm.

Hyperparameters and pseudo-point locations vary among functions in each layer. The length-scales and inducing inputs of the first Gaussian process layer are initialized sensibly based on the median distance between datapoints in the input space and the k-means cluster centers [39], respectively. Longer length-scales and initial inducing inputs between $[-1, 1]$ are used for the higher layers to enforce them to start with an identity mapping. The natural parameters of the average factor are parameterized and initialized with small random values. While the exponentiated quadratic kernel and ARD length-scales are well-suited for this application, alternative kernels such as the Matérn kernel or rational quadratic kernel could also be explored, depending on the specific characteristics of the dataset. For instance, the Matérn kernel, with its adjustable smoothness parameter, might be more appropriate for modeling less smooth or more irregular functions. Similarly, alternative hyperparameter settings, such as different initialization strategies for the length-scales or inducing inputs, could be investigated to further optimize model performance.

4. Results and discussion

4.1. Performance on UCI datasets

DGPAEP was validated to conduct regression experiments on datasets from the University of California, Irvine (UCI) repository. Six

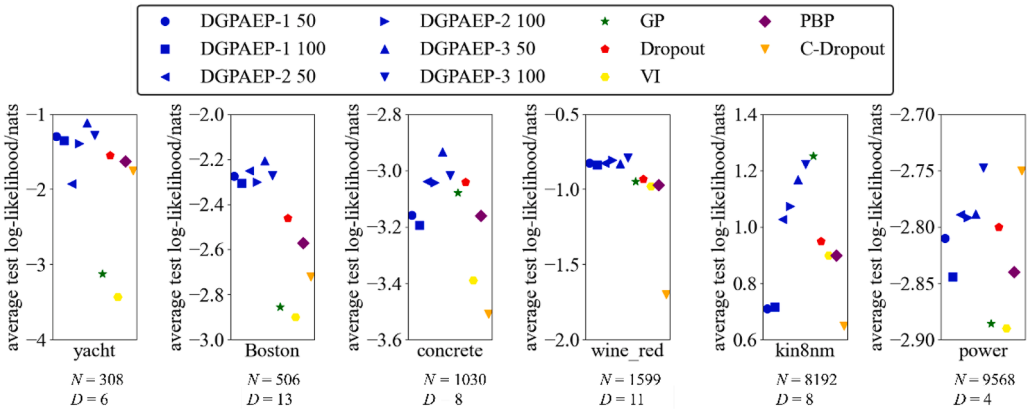


Fig. 8. Average log-likelihood of existing approaches for BNNs and GPs, and the proposed method for DGPs, across 6 datasets. The higher the better, and best viewed in color.

representative datasets were utilized, as detailed (Fig. 8), where MLL was employed as the evaluation metric. It's worth noting that in Fig. 8, N represents the number of samples in the data, and D represents the dimensionality of the features. As N and D increase, they indicate greater data complexity, with their product representing the overall volume of sample data (excluding target values). This metric also reflects data resolution. In this study, we selected datasets of varying sizes—small, medium, and large—to thoroughly assess the performance of the ML model across different dataset scales.

To showcase the high performance of DGPAEP, it was systematically compared with state-of-the-art methods within the Bayesian neural network category, namely Gaussian Process (GP), Dropout, Variational Inference (VI), Probabilistic Backpropagation (PBP), and Concrete Dropout (C-Dropout) [40–44]. These models all inherit the core Bayesian network principle of solving probability distributions between variables. The key difference lies in the methods used for posterior estimation. DGPAEP approximates the marginal likelihood through direct EP energy function approximation to find the approximate posterior distribution over the inducing outputs. The DGPAEP network allows for increasing layers, with each hidden layer having varying numbers of neurons (units). It is important to note that simply changing the number of hidden layers and units per layer does not alter the fundamental estimation method of the network; thus, the improved deep networks still belong to DGPAEP. This study configures three types of hidden layers, $L = 1,2,3$, with each layer having 50 or 100 units, enabling exploration of DGPAEP's predictive performance at different depths to better prepare for predicting CE in subsequent sections. Based on these varying network depths and neuron numbers, we denote these deepened DGPAEP networks as "DGPAEP-L N," where L refers to the number of hidden layers and N indicates the number of neurons per layer. Table 2 shows the parameters set during model training on the public datasets.

Fig. 8 shows the MLL of 6 UCI repository datasets under various predictive models, arranged in descending order. Notably, the blue shape representing DGPAEP attains the top position in 5 out of the 6 datasets, with a slightly lower ranking in kin8nm, yet still within the top three. Furthermore, when compared to the poorest-performing BNN models in each dataset, DGPAEP exhibits a clear visual distinction from these underperforming models, with the models having three layers performing particularly well. Therefore, through testing on public datasets, we initially validate the relative advantages of DGPAEP and select the sub-models with three layers as representatives to further analyze the subsequent research on carbon emission prediction. Table 3 shows the MLL performance of different models on the public datasets, with bolded numbers indicating the highest value in the corresponding dataset. The 3-layer DGPAEP model performs well in terms of numerical analysis. The subsequent analysis will focus on evaluating the model's performance on carbon emission data from WWTP and its ability to enable real-time monitoring.

4.2. Application to carbon emission prediction

Section 3.3 of this study analyzes the feature selection process. From the original 45 × 365 data, 38 variables were selected as features: 11 features in the pretreatment stage, corresponding to numbers 1–11; 18 features in the biological treatment stage, corresponding to numbers 12–29; and 7 features in the sludge treatment stage, corresponding to numbers 30–31 and 33–37. The dataset was divided into training, validation, and test sets with a ratio of 7:1:2. For example, with 365 samples in total, the training set, validation set, and test set contain 255, 37, and 73 samples, respectively. The model training parameters are shown in Table 2.

Table 2
Model training parameter settings in public datasets.

random seed	epochs	points per mini-batch	learning rate	logZ	split	loss function
1234	1000	50	0.02	True	0.2	NLL

Table 3
MLL performance of different models on public datasets.

	yacht	Boston	concrete	wine_red	kin8nm	power
DGPAEP-1 50	−1.30	−2.28	−3.16	−0.82	0.71	−2.81
DGPAEP-1 100	−1.35	−2.31	−3.20	−0.84	0.20	−2.84
DGPAEP-2 50	−1.93	−2.25	−3.04	−0.82	1.03	−2.79
DGPAEP-2 100	−1.39	−2.30	−3.04	−0.81	1.07	−2.79
DGPAEP-3 50	**−1.12**	**−2.21**	**−2.93**	−0.83	1.17	−2.79
DGPAEP-3 100	−1.28	−2.27	−3.02	**−0.79**	1.22	**−2.74**
GP	−3.12	−2.85	−3.08	−0.95	**1.25**	−2.89
Dropout	−1.55	−2.46	−3.04	−0.93	0.95	−2.80
VI	−3.43	−2.90	−3.39	−0.98	0.90	−2.89
PBP	−1.63	−2.57	−3.16	−0.97	0.90	−2.84
C-Dropout	−1.75	−2.72	−3.51	−1.70	0.65	−2.75

Based on the results from testing on public datasets, as shown in Fig. 8, DGPAEP exhibits significant improvements over the previous state-of-the-art Bayesian models. To validate whether DGPAEP maintains high generalization capabilities in the carbon emission domain, we selected other relatively well-performing ML models from public datasets for comparison against our model on specific datasets. Thus, for predicting CE in WWTP, we selected four models: the traditional GP model, C-Dropout, and our proposed DGPAEP sub-models, DGPAEP-3 50 and DGPAEP-3 100. To facilitate comprehension, the colors in the bar charts correspond to those in Fig. 8. Additionally, to differentiate between DGPAEP-3 50 and DGPAEP-3 100, we used different shading patterns in the respective bar charts for intuitive distinction. MLL evaluates the model's fit, with higher values indicating stronger explanatory power. As depicted in Fig. 9(a), DGPAEP consistently demonstrates superior fitting performance. Although DGPAEP-3 100 shows better fitting in the first two stages compared to DGPAEP-3 50, the latter excels in the third stage, significantly outperforming the GP model in second place. From the MLL results, our improved model shows a 149.07 % enhancement compared to the baseline model, which is the GP model used in the other Bayesian neural network models (as shown in Fig. 9). RMSE assesses the prediction error, measuring the difference between predicted and actual values. Lower RMSE values indicate predictions closer to the true values. As shown in Fig. 9(b), DGPAEP consistently achieves better RMSE results compared to other models, with particularly notable advantages. In the sludge treatment stage, although DGPAEP-3 50 underperforms, DGPAEP-3 100 still surpasses the other models. Across the three stages, DGPAEP-3 100 achieves an RMSE of 8.53, reflecting a 3.05 % improvement over the GP model and other state-of-the-art Bayesian models. Therefore, both in terms of MLL, which emphasizes model fit, and RMSE, which emphasizes accuracy, DGPAEP demonstrates excellent performance. To address potential overfitting concerns, the model incorporates sparse approximation and hierarchical structure techniques that help mitigate overfitting risks. Firstly, sparse approximation methods, such as FITC and EP, significantly reduce model complexity by introducing inducing points. These points represent a subset of the data, allowing the model to approximate the full Gaussian process without the computational burden and potential overfitting that would arise from using all data points. By limiting the number of inducing points, the model generalizes better to unseen data, focusing only on the most relevant information while avoiding overfitting to noise or irrelevant details in the data. Additionally, the hierarchical structure in the DGPAEP architecture further addresses overfitting by structuring the model in multiple layers. Each layer captures different levels of data abstraction, with higher layers focusing on more complex patterns while lower layers model simpler relationships. This regularization effect ensures that the model does not rely too heavily on any one layer, thus avoiding overfitting to specific features or noise in the data. The smaller number of inducing points used in the

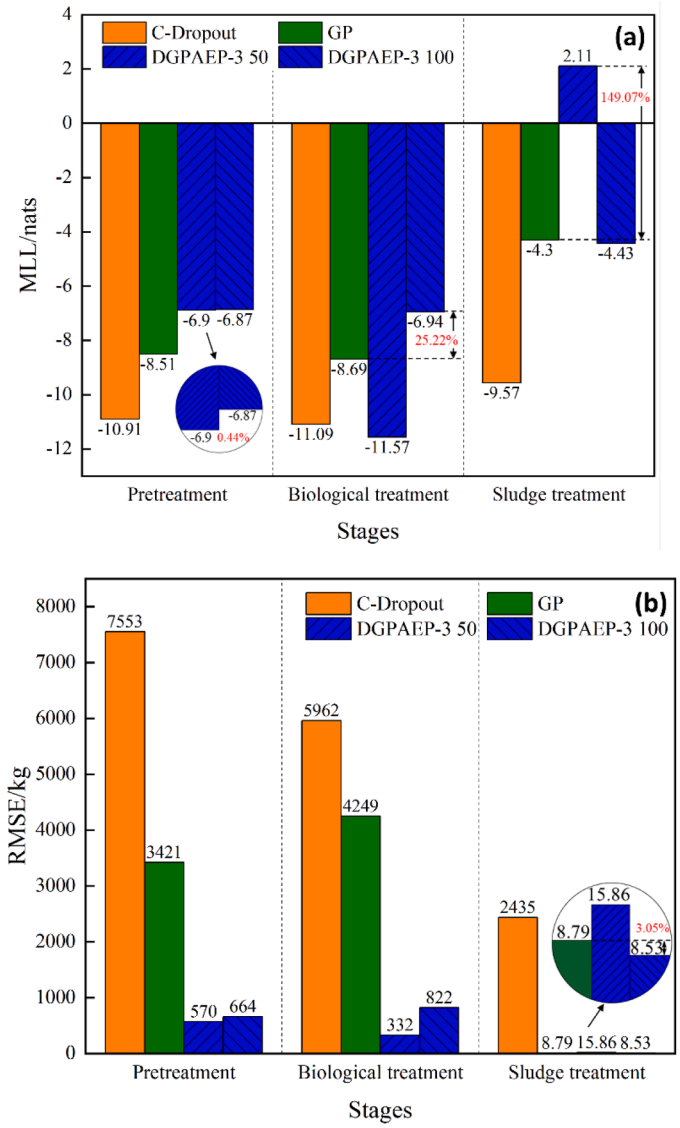


Fig. 9. Comparison of model performance metrics MLL and RMSE across three stages: (a) MLL, (b) RMSE.

higher layers also acts as a form of regularization, further limiting the model's capacity to memorize data and encouraging better generalization. Together, these techniques—sparse approximation and hierarchical modeling—ensure that the model remains robust and generalizable, addressing the potential overfitting issues while maintaining high predictive performance. These experimental results suggest that DGPAEP enhances predictive capabilities and is well-suited for the carbon emission domain, indicating that it can be specifically applied to this field. Table 4 shows in detail the performance metrics of the DGPAEP model in this study, and the bolded parts are the optimal values in the comparison experiments.

In the previous sections, we have visually demonstrated the significant volatility and uncertainty of CE. Experiments have shown that the improved model presented in this paper exhibits better performance, marking a significant advancement in the realm of ML. We now visualize the predictions generated by the DGPAEP model, as shown in Fig. 10. Fig. 10(a), 10(b), and 10(c) represent the pretreatment stage, biological treatment stage, and sludge treatment stage, respectively. The x-axis of the scatter plots represents the true values, while the y-axis represents the predicted values. The blue solid diagonal line, based on the principle of a linear function ($y = x$), represents the line of ideal predictions, where the predicted values exactly match the true values. The red and green scatter points denote the DGPAEP predictions, with points closer to the diagonal line indicating better predictive performance. To provide a more rigorous assessment of the model's accuracy, we calculated the coefficient of determination (R^2), which measures the proportion of variance in the true values explained by the model. The (R^2) value is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y}_i)^2} \tag{14}$$

where y_i is the true value, $\hat{y}_i$ is the predicted value, and $\bar{y}$ is the mean of the true values. The R^2 values for the DGPAEP model are displayed in Fig. 11, demonstrating strong agreement between the predicted and true values across all stages. Additionally, the RMSE was computed to further quantify the model's performance, as shown in Fig. 9(b). To enhance clarity and display more details, we have zoomed in on key areas within the scatter plots. These zoomed-in regions focus on data-dense areas where the differences between predicted and true values are subtle and difficult to discern at full scale. By magnifying these regions, we aim to provide a clearer visualization of the model's predictive accuracy, particularly in areas where data points are densely clustered. This approach allows for a more intuitive understanding of the discrepancies between predicted and true values, rather than simply overlapping points.

For a comprehensive comparison, Fig. 12 presents line plots depicting the predictions of all models included in the study (DGPAEP, GP, and C-Dropout) on the same axes. This side-by-side visualization enables a direct assessment of DGPAEP's performance relative to the alternatives. The RMSE values for each model are also displayed in Fig. 9(b), providing a quantitative measure of their predictive accuracy. The line plots illustrate the predicted trends of each model, showing how well they capture the variations in the actual data. The DGPAEP model demonstrates a closer fit to the true values compared to the other models, particularly in capturing fluctuations and extreme values. Although some deviations occur at the peaks and troughs, the overall predictive accuracy remains high, as evidenced by the superior RMSE.

After completing the model training, we can now achieve high-precision predictions of CE with the help of ML. However, this does not fully meet the objectives of this study. Beyond accurate predictions, it is essential to understand the extent to which various factors influence carbon emissions, as this knowledge is crucial for developing effective emission reduction strategies. SHAP, based on Shapley values from cooperative game theory [45], provides a unified framework for interpreting the contributions of individual features to the model's predictions. Fig. 13 illustrates the feature importance of variables for the DGPAEP model across the three stages of wastewater treatment. The horizontal axis represents the mean SHAP value, where a larger value

Table 4
Average RMSE and Log-Likelihood for the selected models across three stages.

Datasets	Avg. RMSE				Avg. LL			
	C-Dropout	GP	DGPAEP-3 50	DGPAEP-3 100	C-Dropout	GP	DGPAEP-3 50	DGPAEP-3 100
Pretreatment	7553.01	3421.03	**569.99**	663.75	−10.91	−8.51	−6.90	**−6.87**
Biological treatment	5961.61	4249.03	**331.99**	822.19	−11.09	−8.69	−11.57	**−6.94**
Sludge treatment	2434.77	8.79	15.86	**8.53**	−9.57	−4.30	**2.11**	−4.43

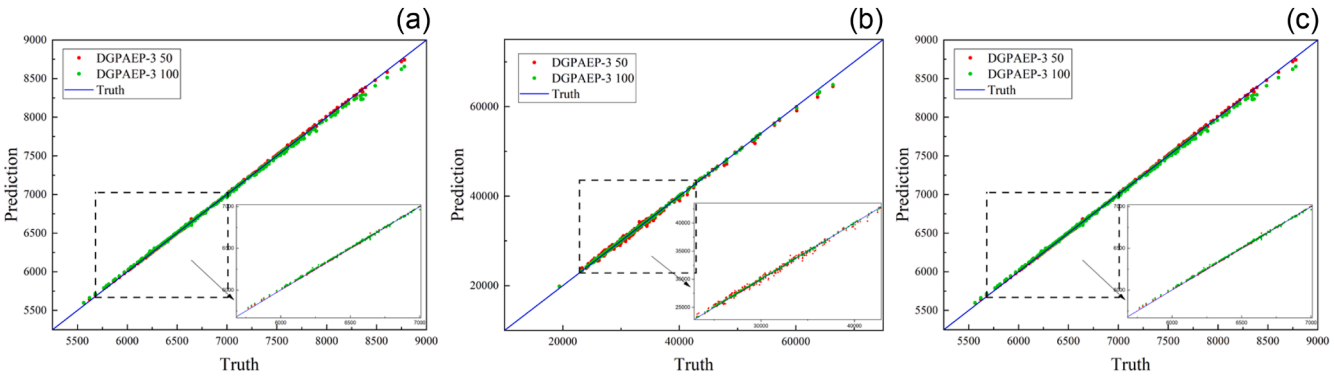


Fig. 10. Scatter plots of predicted vs. true carbon emissions for different models across three stages: (a) Pretreatment stage, (b) Biological treatment stage, (c) Sludge treatment stage. The blue solid line represents the ideal actual values, and the closer the scatter points are to this line, the more accurate the predictions. Key areas in the plots have been locally magnified for better observation.

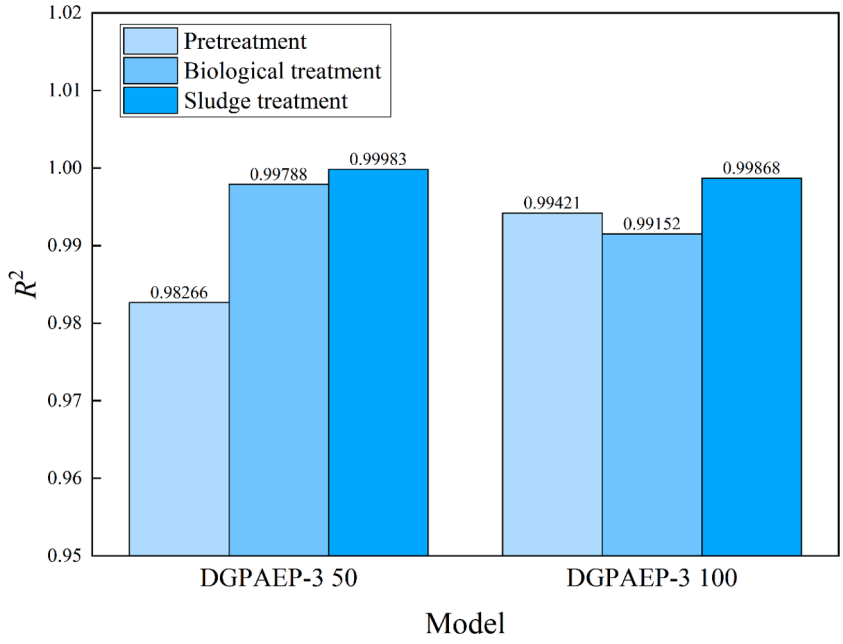


Fig. 11. R^2 values for the DGPAEP model, demonstrating strong agreement between the predicted and true values across all stages.

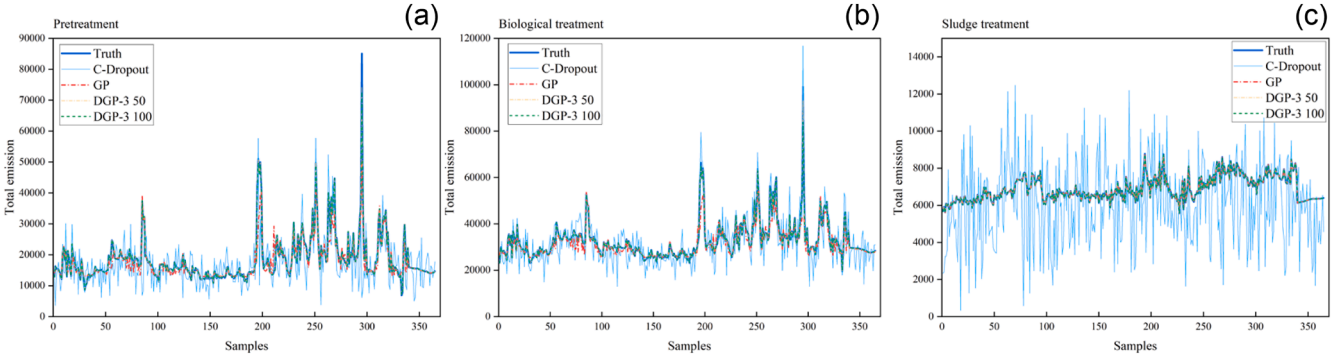


Fig. 12. Line plots showing the predictions of all models (DGPAEP, GP, and C-Dropout) on the same axes for comparison.

indicates a greater impact on the model's final output. Features are ranked in descending order of importance. In Fig. 13(a), the dosage of Ferric Chloride ($FeCl_3$) in the primary sedimentation tank ($FeCl_3$-PST) emerges as the most influential factor, followed closely by the dosage of Ferric Sulfate ($Fe_2(SO_4)_3$) Polymer in the same stage ($Fe_2(SO_4)_3$-PST), both exhibiting nearly identical contributions. While identifying these key features is valuable, it is equally important to analyze how adjusting these factors influences emissions and whether their effects are linear, threshold-based, or dependent on interactions with other variables.

For instance, reducing $FeCl_3$ dosage may lower direct emissions from

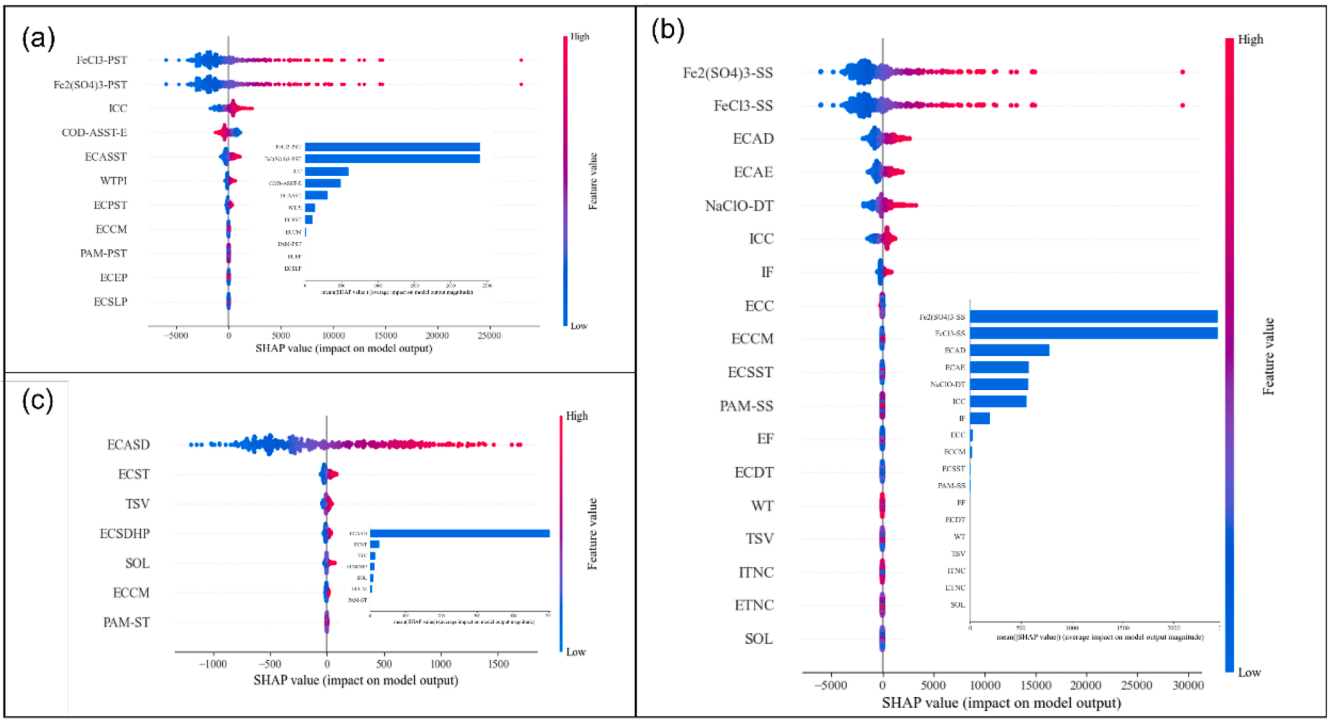


Fig. 13. SHAP values for each feature's impact on the predicted values of the DGPAEP model across three stages: (a) Pretreatment stage, (b) Biological treatment stage, and (c) Sludge treatment stage. Higher SHAP values indicate a greater influence on the prediction results, as shown in the scatter plots. To enrich the information, a horizontal bar chart is inserted in the lower right corner of each scatter plot, representing the mean SHAP values for each feature; larger values indicate a greater impact on the predictions.

chemical consumption, but the effect is unlikely to be strictly proportional. A reduction beyond a certain threshold could compromise coagulation efficiency, leading to higher organic loads in subsequent treatment stages. This, in turn, could increase aeration demand in the biological treatment phase, thereby raising indirect emissions from electricity consumption. Similarly, an insufficient dosage of $Fe_2(SO_4)_3$ may lead to incomplete phosphorus removal, requiring additional chemical inputs downstream. These findings suggest that optimizing chemical dosing must consider trade-offs between direct and indirect emissions rather than focusing solely on reducing individual chemical inputs. Moreover, emissions in WWTPs are rarely influenced by a single factor in isolation. Instead, complex interactions exist between multiple process parameters. The SHAP analysis reveals the individual contributions of each feature, but a deeper investigation into feature interactions is necessary to uncover combined effects. For example, the impact of $FeCl_3$ dosage may vary depending on influent COD concentration, aeration rate, and pH levels. Higher COD levels could increase the demand for $FeCl_3$ due to higher particulate matter in the influent, while variations in aeration intensity may alter the effectiveness of $FeCl_3$ in settling suspended solids. Additionally, interactions between carbon source dosage (CSD), nitrogen removal efficiency (TN-eff), and dissolved oxygen levels (DO) could provide further insights into the interdependencies of chemical dosing, biological treatment efficiency, and aeration energy consumption. To enhance the interpretability of these findings, a horizontal bar chart is included at the bottom right of Fig. 13, displaying the average contribution of each feature to the output. The ranking in the bar chart aligns with that in the scatter plot, reinforcing the consistency of feature importance across different visualization methods. By integrating feature importance analysis with an understanding of feature interactions, these insights can support more targeted interventions. Future work should incorporate interaction analysis techniques, such as SHAP interaction values or partial dependence plots, to quantify how combined effects influence emissions. This approach will enable wastewater treatment plants to develop dynamic optimization strategies that balance emission reduction with process efficiency, ultimately contributing to more sustainable operations.

The SHAP dependence plot is a visualization tool provided by SHAP to illustrate how a feature's SHAP value—its contribution to the model's predictions—varies with its actual value. By analyzing SHAP dependence plots, researchers can gain insights into whether a feature's impact on the model output is linear or nonlinear, whether a threshold effect exists, and whether interactions with other features influence its effect. Fig. 14, Fig. 15, and Fig. 16 respectively represent SHAP dependence plots for the pretreatment, biological treatment, and sludge treatment stages. These plots help uncover relationships between features and model outputs. A SHAP dependence plot typically appears as a scatter plot, with the x-axis representing the feature's actual values and the y-axis indicating its SHAP values, reflecting its impact on model predictions. The color of each point denotes the most strongly interacting feature. For instance, in the biological treatment stage, the effect of ECAD is closely related to ECAE, as shown in the plot where ECAE values are color-coded. The plot clearly indicates a positive linear correlation between ECAE and ECAD, meaning that as ECAE increases, carbon emissions also rise. However, $Fe_2(SO_4)_3$-SS has an inverse relationship with ECC, where lower $Fe_2(SO_4)_3$-SS levels contribute to higher carbon emissions. Furthermore, SHAP dependence plots reveal feature interactions. A noticeable shift in scatter point colors suggests that a feature's influence is modulated by another feature. For example, the impact of $FeCl_3$ dosage is relatively minor at low aeration rates but becomes significantly stronger under high aeration conditions.

4.3. *Challenges in real-world applications*

Despite the promising results demonstrated by DGPAEP, several challenges remain when applying the model to real-world WWTPs. These challenges must be addressed to ensure the model's robustness, scalability, and practical feasibility. One major challenge lies in data acquisition and quality. Carbon emission data from WWTPs may be

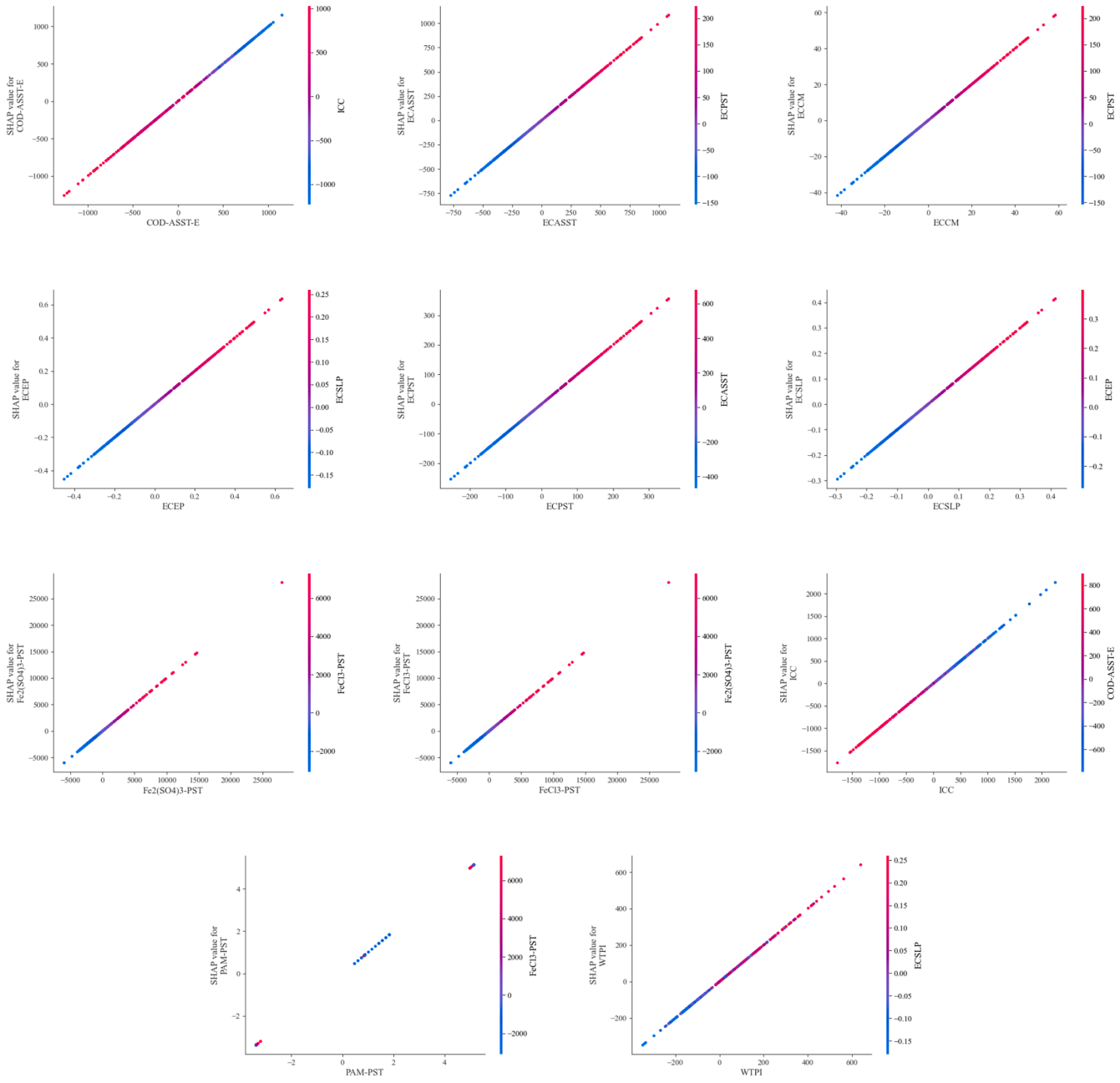


Fig. 14. SHAP dependence plots of 11 features in the pretreatment stage.

affected by sensor inaccuracies, missing values, and manual entry errors, all of which can introduce noise and inconsistencies into the dataset. Although the data used in this study underwent rigorous preprocessing to mitigate these issues, the broader application of the model across different facilities may still be constrained by data heterogeneity and limited availability. Future research could explore advanced data imputation techniques and anomaly detection methods to enhance data reliability. Another critical concern is model generalization. While DGPAEP demonstrated superior performance on both the WWTP dataset and UCI benchmark datasets, variations in treatment processes, climatic conditions, and plant infrastructure across different WWTPs may impact its adaptability. To enhance generalization, future studies could integrate transfer learning or domain adaptation techniques, enabling the model to adjust dynamically to new operating environments without extensive retraining.

Furthermore, computational resource requirements present a significant barrier to practical deployment. Bayesian neural networks, particularly those employing deep Gaussian processes, involve substantial computational overhead, which may be prohibitive for real-time inference in WWTPs that lack high-performance computing infrastructure. Efficient model compression techniques, such as knowledge distillation and parameter pruning, could be explored to reduce computational complexity while maintaining predictive accuracy. Alternatively, cloud-based or edge computing solutions could facilitate real-time deployment with lower hardware demands. Another aspect that requires attention is temporal variability and dynamic changes in carbon emissions. Seasonal and periodic fluctuations in wastewater composition and treatment efficiency can influence emission patterns, potentially limiting the model's long-term predictive accuracy. While DGPAEP implicitly captures these variations through data-driven learning, incorporating time-series forecasting architectures, such as recurrent neural networks (RNNs), LSTM networks, or Transformer-

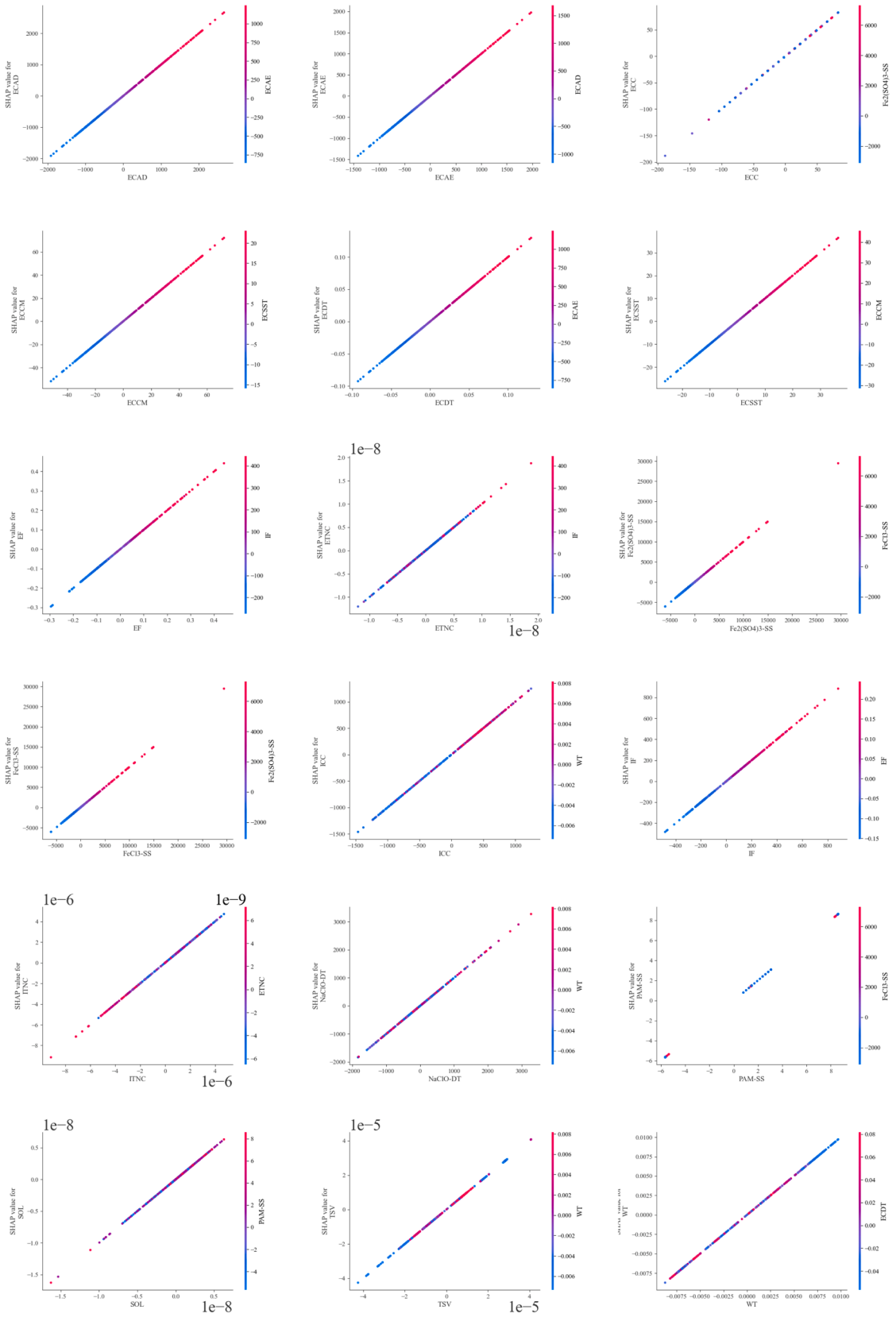


Fig. 15. SHAP dependence plots of 18 features in the biological treatment stage.

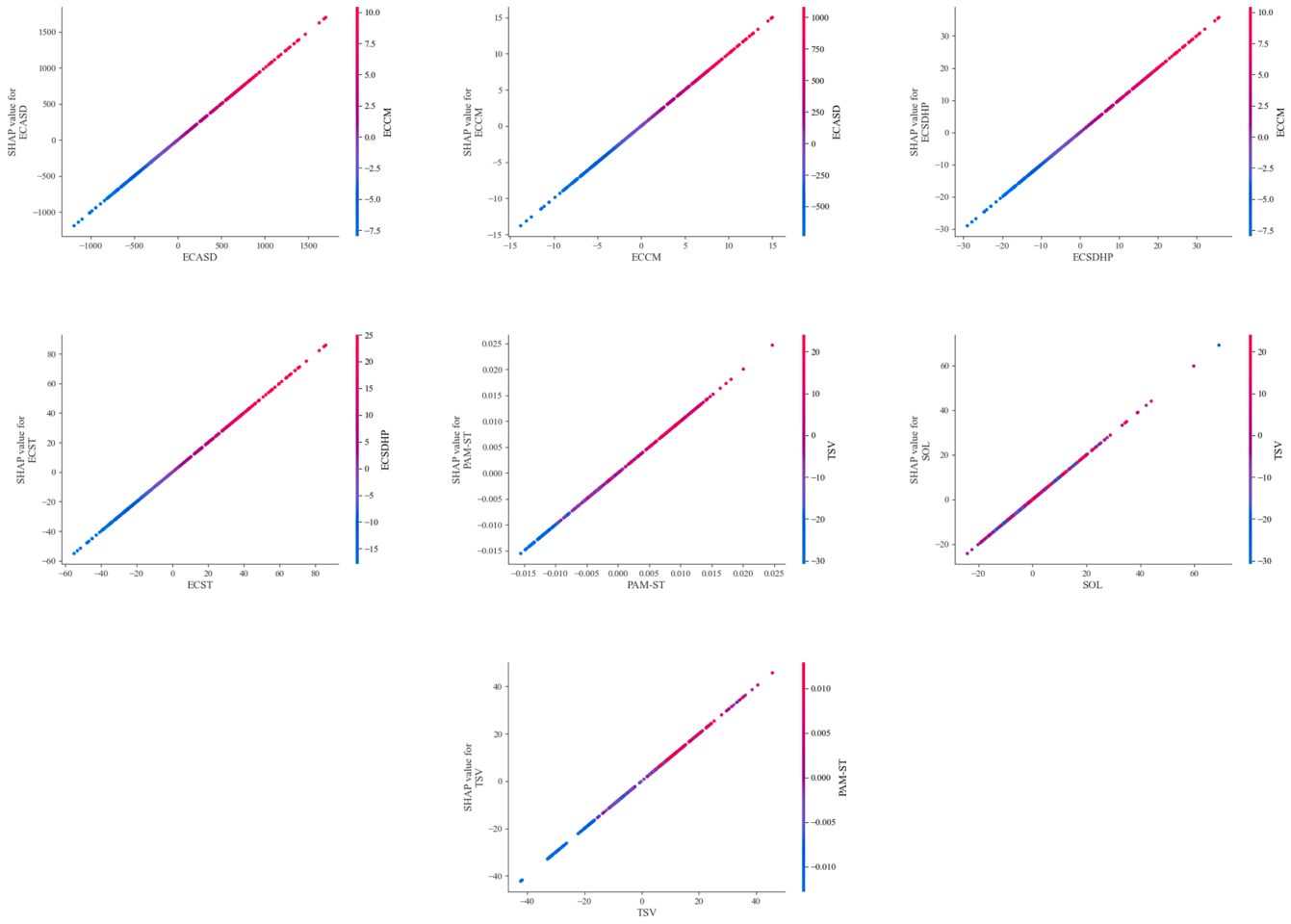


Fig. 16. SHAP dependence plots of 7 features in the sludge treatment stage.

based models, could further improve its adaptability to dynamic operational conditions.

Lastly, model interpretability and acceptance remain essential considerations for real-world adoption. Although SHAP-based explainability enhances the interpretability of model predictions, WWTP operators may require additional decision-support tools to translate insights into actionable strategies. Future work could focus on developing interactive visualization dashboards that integrate SHAP analysis with real-time process monitoring, thereby facilitating informed decision-making for carbon emission reduction. Addressing these challenges will be crucial for the widespread adoption of DGPAEP in real-world wastewater treatment applications. By integrating robust data handling techniques, enhancing model adaptability, optimizing computational efficiency, and improving interpretability, future research can further refine DGPAEP's capabilities, ultimately contributing to more sustainable and efficient carbon emission management in WWTPs.

5. Conclusions

This study focuses on improving a Bayesian neural network-based ML model and successfully applying it to the CE domain, using a full-scale wastewater treatment plant as the research object. Although previous literature has seen the application of enhanced ML models for carbon emission prediction, these typically involve simple model integration, leaving their generalization capabilities uncertain. To address this, we refined the Gaussian process model by adopting a novel and robust deterministic approximation scheme, employing an approximate EP algorithm and a FITC approximation to overcome computational and analytical challenges. To tackle the marginalization problem inherent in the approximate EP algorithm, a new probabilistic backpropagation algorithm extension was developed, significantly enhancing the model's generalization capability and prediction accuracy. Experimental results demonstrate that on public datasets of varying sizes, the DGPAEP consistently outperforms other state-of-the-art Bayesian network models in terms of MLL, proving its strong generalization capability. Subsequently, the model was applied to the CE field, yielding impressive results, with MLL improving by 149.07 % and RMSE decreasing by 3.05 %. Notably, this study utilized SHAP values to quantify the contributions of different features in the data to the DGPAEP's predictions, thereby providing scientific guidance for emission reduction efforts.

CRediT authorship contribution statement

Yonglin Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Yongmu Li:** Validation, Supervision, Resources, Project administration, Investigation. **Kun Zheng:** Writing – review & editing, Supervision, Resources, Project administration, Investigation. **Chao Meng:** Writing – review & editing, Project administration, Investigation, Funding acquisition. **Changjian Zhu:** Writing – review & editing, Supervision, Project administration. **Zenghai Shan:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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